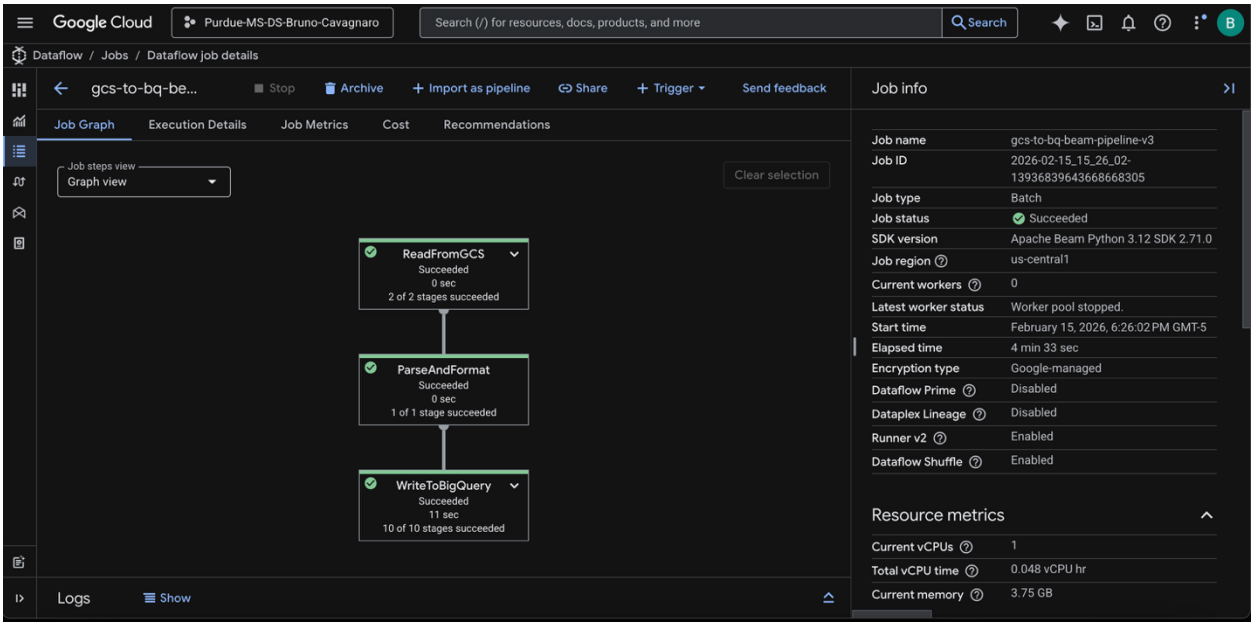
Laboratory 2:

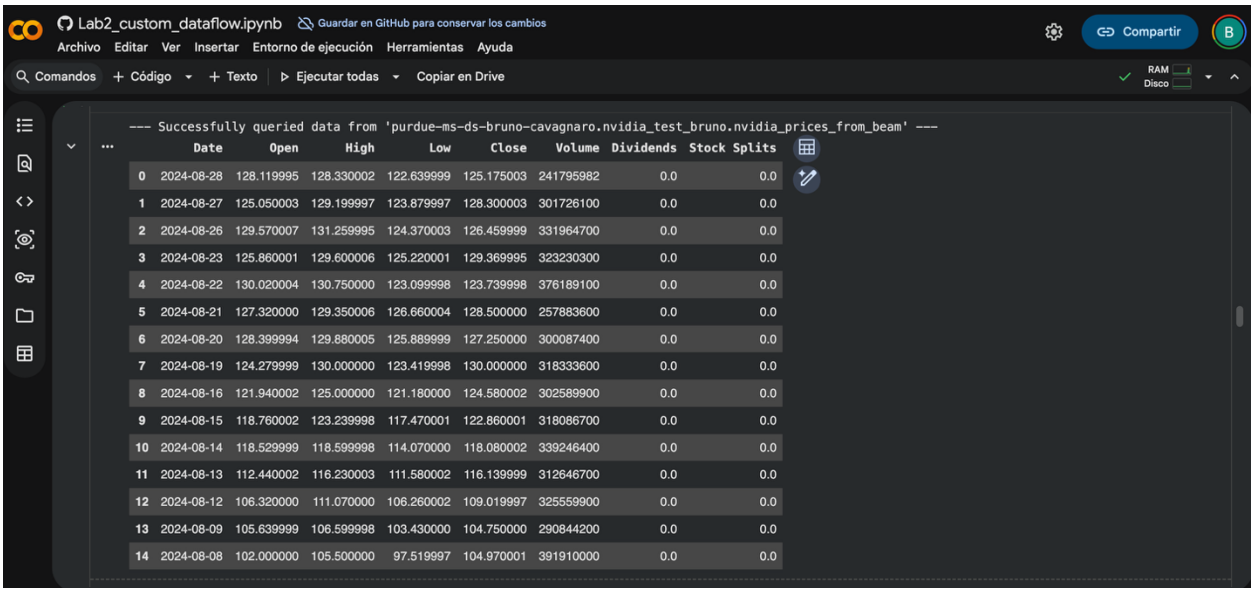
Link to the jupyter notebook, with all the code:

https://github.com/BrunoCavag/Alicorp/blob/master/Lab2_custom_dataflow.ipynb



List of the different actions:

1. Cell 1: Install Required Libraries and Resolve Dependencies
2. Cell 2: Authenticate and Configure Project Variables (in this cell I have modified the different parameters of temporal buckets, bigquery dataset, etc; and the table schema, because is a different dataset).
3. Cell 3: Write the Custom Apache Beam Pipeline Script (here a modified the parse script, so It matches the schema of this new dataset)
4. Cell 4: Launch the Dataflow Job
5. Cell 5: Verify the Final Result in BigQuery



	Date	Open	High	Low	Close	Volume	Dividends	Stock Splits
0	2024-08-28	128.119995	128.330002	122.639999	125.175003	241795982	0.0	0.0
1	2024-08-27	125.050003	129.199997	123.879997	128.300003	301726100	0.0	0.0
2	2024-08-26	129.570007	131.259995	124.370003	126.459999	331964700	0.0	0.0
3	2024-08-23	125.860001	129.600006	125.220001	129.369995	323230300	0.0	0.0
4	2024-08-22	130.020004	130.750000	123.099998	123.739998	376189100	0.0	0.0
5	2024-08-21	127.320000	129.350006	126.660004	128.500000	257883600	0.0	0.0
6	2024-08-20	128.399994	129.880005	125.889999	127.250000	300087400	0.0	0.0
7	2024-08-19	124.279999	130.000000	123.419998	130.000000	318333600	0.0	0.0
8	2024-08-16	121.940002	125.000000	121.180000	124.580002	302589900	0.0	0.0
9	2024-08-15	118.760002	123.239998	117.470001	122.860001	318086700	0.0	0.0
10	2024-08-14	118.529999	118.599998	114.070000	118.080002	339246400	0.0	0.0
11	2024-08-13	112.440002	116.230003	111.580002	116.139999	312646700	0.0	0.0
12	2024-08-12	106.320000	111.070000	106.260002	109.019997	325559900	0.0	0.0
13	2024-08-09	105.639999	106.599998	103.430000	104.750000	290844200	0.0	0.0
14	2024-08-08	102.000000	105.500000	97.519997	104.970001	391910000	0.0	0.0

6. Linear Regression (Predicting a Stock's High Price)
7. Logistic Regression (Predicting a High Volume Day)
8. K-Means Clustering (Grouping Similar Trading Days)